

Building M4350 Binary from Fastpath M4350 Source code

1. Download and Create Virtual Machine

- Download Rocky Linux 8.10 from <https://rockylinux.org/download>
- Create a virtual machine with a 100GB disk and dynamic RAM allocation.

2. Install EPEL-Release and Development Tools

- Run the following commands to install EPEL, enable powertools, and update your system:

```
dnf -y install epel-release
dnf config-manager --set-enabled powertools
dnf -y groupinstall "Development Tools"
dnf -y update
```

3. Install Required Packages with yum

- Use yum to install the following packages:
 - Make
 - Zlib-devel.i686
 - libgcc
 - libgcc.i686
 - glibc-devel
 - glibc-devel.i686
 - libstdc++-devel
 - elfutils-libelf-devel
 - m4
 - readline-devel
 - openssl-devel
 - cpan
 - rpm-build
 - swig
 - flex
 - bison
 - gcc

- autoconf
- lzma
- dtc
- libstdc++-static
- glibc-static
- glibc-static.i686

For example:

```
yum -y install make zlib-devel.i686 libgcc libgcc.i686 glibc-devel glibc-devel.i686
libstdc++-devel elfutils-libelf-devel m4 readline-devel openssl-devel cpan
rpm-build swig flex bison gcc autoconf lzma dtc libstdc++-static glibc-static
glibc-static.i686
```

4. Install 7za

- Download 7za from
<https://sourceforge.net/projects/p7zip/files/p7zip/9.20.1/>
- Extract and install it:

```
tar jxf p7zip_9.20.1_x86_linux_bin.tar.bz2
cd p7zip_9.20.1
./install.sh
```

5. Install Doxygen and Java

- Install Doxygen and OpenJDK 1.8:

```
dnf -y install doxygen doxygen-latex doxygen-doxywizard
dnf -y install java-1.8.0-openjdk-devel
```

6. Install and Set Up Software Tools

- Install PGen and Python2, set up symlinks, and prepare the build environment:

```
pip3 install PGen
yum -y install python2
sudo alternatives --set python /usr/bin/python2
mkdir -p /tools/bin
ln -s /bin/python2 /tools/bin/python
wget https://www.python.org/ftp/python/2.7.18/Python-2.7.18.tgz
tar xvfz Python-2.7.18.tgz
```

```

cd Python-2.7.18
./configure --prefix=/tools
sudo dnf config-manager --set-enabled powertools
make python Parser/pgen
cp Parser/pgen /tools/bin/pgen
chmod 777 /tools/bin/pgen
ln -sf /usr/bin/python /tools/bin/python
ln -sf /usr/bin/install /tools/bin/install
cpan Shell
cpan JSON
yum -y install "perl(Tie::IxHash)"
ln -s /usr/sbin/mksquashfs /usr/sbin/mksquashfs4.2

```

- If python --version does not show Python 2.7.18, use:

```

sudo alternatives --install /usr/bin/unversioned-python python /usr/bin/python2 1
sudo alternatives --set python /usr/bin/python2
sudo ln -sf /usr/bin/python2 /usr/bin/python

```

7. Install Lua 5.2

- Download and build Lua 5.2 from <https://www.lua.org/ftp/>

```

tar xvfz lua-5.2.4.tar.gz
cd lua-5.2.4
make linux
cp src/lua /usr/bin/lua52

```

8. Prepare Toolchain

- Move the toolchain archive (**FASTPATH_R8131_toolchain.tar.bz2**) to the root and extract it:

```

cd /
tar xvjf FASTPATH_R8131_toolchain.tar.bz2

```

9. Set Library Environment Variable

- Set the **LD_LIBRARY_PATH** where you intend to use the toolchain:

```

export LD_LIBRARY_PATH=/projects/nwsoft-toolchains/xldk/iproc-a72-d42a6432

```

10. Build mkimage Tools

- Create a netgear directory in admin directory, extract open source support code, and build mkimage tools:

```
mkdir -p /home/admin/netgear
cd /home/admin/netgear
tar xvjf FASTPATH_R8131_open_source_support_code.tar.bz2
cd
/home/admin/netgear/FASTPATH_R8131_open_source_support_code/u-boot-2012.10
make tools
cp tools/mkimage tools/mkimage64
mkdir -p /tools/nwsoft/lib/
cp tools/mkimage tools/mkimage64 /tools/nwsoft/lib/

cd /home/admin/netgear/FASTPATH_R8131_open_source_support_code/u-boot-arm64
export ARCH=arm64
export
CROSS_COMPILE=/projects/nwsoft-toolchains/xldk/iproc-a72-d42a6432/bin/aarch64-b
uildroot-linux-gnu-
make mrproper
make northstar2_defconfig
make tools
cp tools/mkimage tools/mkimagearm64
cp tools/mkimagearm64 /tools/nwsoft/lib/mkimagearm64
```

11. Build Linux Kernel

- To know the kernel configuration file used for a platform, inspect the bsp/cpu/<cpu name>/<kernel name>/kernel.cfg file. L7_LINUX_CONFIG_FILE specifies the name of the kernel configuration file.
- In M4350 the kernel configuration file is bcm_ns2_m4350_defconfig and the path is /home/admin/netgear/source_code/bsp/cpu/m4350_ns2/ldk_6.x
- Build kernel as follows:

```
cd /home/admin/netgear/FASTPATH_R8131_open_source_support_code/fp_ldk_6.x
ARCH=arm64
CROSS_COMPILE=/projects/nwsoft-toolchains/xldk/iproc-a72-d42a6432/bin/aarch64-b
uildroot-linux-gnu- make bcm_ns2_m4350_defconfig
```

ARCH=arm64

CROSS_COMPILE=/projects/nwsoft-toolchains/xldk/iproc-a72-d42a6432/bin/aarch64-b
uildroot-linux-gnu- make

12. Prepare Source Code Directory and Configuration

- Please download the source code file (M4350_14.0.6.13-src.tar.bz2) from Broadcom ESP portal.
- Create the source_code directory and extract your package:

```
mkdir -p /home/admin/netgear/source_code  
cd /home/admin/netgear/source_code  
tar xvjf M4350_14.0.6.13-src.tar.bz2
```

- Edit the following files as described:

a. **bsp/cpu/common/linux/ipl/makerules.link** (replace these lines as below):

```
vim bsp/cpu/common/linux/ipl/makerules.link
```

- Replace:
\$(DBG) \$(LZMA_LEGACY_CMD) e \$(patsubst %.lzma,%,\$@) \$@ >/dev/null
2>&1
- with
\$(DBG) \$(LZMA_LEGACY_CMD) \$(patsubst %.lzma,%,\$@) >/dev/null 2>&1
- Replace:
\$(DBG) \$(LZMA_LEGACY_CMD) e \$(out_system)/rootfs_image
\$(out_system)/rootfs_image.lzma>/dev/null 2>&1
- with
\$(DBG) \$(LZMA_LEGACY_CMD) \$(out_system)/rootfs_image >/dev/null 2>&1

b. **make_configure file**

```
vim make_configure
```

- Update these variables and leave the OUTPATH line as it is:

```
export WORK_DIR := /home/admin/netgear/  
export REPOS_HOME := $(WORK_DIR)/source_code/  
export FP_BASE := $(REPOS_HOME)  
export CROSS_COMPILE:=  
/projects/nwsoft-toolchains/xldk/iproc-a72-d42a6432/bin/aarch64-buildroot-linux-gnu-  
export LD_LIBRARY_PATH=$LD_LIBRARY_PATH:/tools/nwsoft/lib  
export PERL5LIB := .:$PERL5LIB  
export PATH := /usr/local/bin:$(PATH)
```

c. **src/mgmt/snmp/make_mibs/processmib.pl** (fix regex for braces)

```
vim src/mgmt/snmp/make_mibs/processmib.pl
```

- Replace:
my \$line = \$_ if /^[:blank:]]*::= { fastPath/;
- with:
my \$line = \$_ if /^[:blank:]]*::= \{ fastPath/;
- Replace:
my \$line = \$_ if /^[:blank:]]*::= { agentDot1qMrpMxrp/;
- with:
my \$line = \$_ if /^[:blank:]]*::= \{ agentDot1qMrpMxrp/;

13. Set Kernel Source path and Build the Fastpath

- Export the kernel source path:

```
export  
KERNEL_SRC=/home/admin/netgear/FASTPATH_R8131_open_source_support_code/  
fp_ldk_6.x
```

- Build with:

```
make
```

14. M4350 binary stk file

- M4350 binary stk file (M4350-v14.0.6.13.stk) is generated at the following path:

```
/home/admin/netgear/source_code/output/Cheetah-M4350Series-Ent-esw-m4350-m4350_ns2-IL6XR-CTxw-6AIQHS/ipl
```

15. Changing the build number

- Open the following file

```
/home/admin/netgear/source_code/output/Cheetah-M4350Series-Ent-esw-m4350-m4350_ns2-IL6XR-CTxw-6AIQHS/package.cfg
```

You will find the following lines. Please edit them to change the Build number as you need.

```
export L7_REL_NUM=14
export L7_VER_NUM=0
export L7_MAINT_NUM=6
export L7_BUILD_NUM=13
```